

Variational Autoencoder-based Analysis of Respiratory Sounds and Latent-space Regression for Severity Estimation

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Abstract—Respiratory diseases frequently manifest through abnormal acoustic patterns such as crackles and wheezes. Although auscultation is widely used, its interpretation can be subjective and sensitive to noise and recording conditions. This study proposes a deep learning framework that uses a Variational Autoencoder (VAE) to learn compact latent representations of lung sound Mel-spectrograms and subsequently quantifies lung-condition severity via regression in the latent space. Audio recordings are standardized through a reproducible preprocessing pipeline and transformed into normalized 64×64 Mel-spectrograms. After VAE training, the encoder extracts informative latent vectors for each sample, and a regression model maps these vectors to a continuous severity score expressed as the percentage of normal lung function.

Index Terms—Respiratory sound analysis, variational autoencoder, Mel-spectrogram, latent representation, severity estimation, regression.

I. INTRODUCTION

Respiratory disorders such as asthma, COPD, and pneumonia are often associated with abnormal lung sounds, including crackles and wheezes. In clinical settings, auscultation remains a standard practice; however, manual interpretation is subjective and can vary across clinicians and environments. As a result, automated analysis of respiratory acoustics has become an important research direction for scalable screening and decision support.

This work builds a VAE-based framework that learns structured latent representations from lung sound spectrograms and then uses those representations to estimate a continuous measure of severity. The project emphasizes: (i) robust data preprocessing from raw recordings and annotations, (ii) Mel-spectrogram generation and normalization for neural learning, (iii) VAE-based latent feature learning, and (iv) regression on latent vectors to estimate severity as a percentage of normal lung function.

II. LITERATURE REVIEW

Automated respiratory sound analysis commonly faces two practical constraints: limited labeled data (often imbalanced across pathologies) and high heterogeneity in recordings due to

stethoscope type, sampling rate, background noise, and patient variability. These constraints motivate approaches that can both learn robust representations and improve data efficiency through augmentation and generative modeling [1].

Spectrogram-based deep learning has shown strong promise for respiratory sound classification in realistic clinical settings. Deep learning models trained on time–frequency representations can discriminate respiratory sound types (including crackles and wheezes) effectively, supporting the feasibility of real-world deployment when careful preprocessing is used [2]. This finding reinforces the role of Mel-spectrogram and MFCC-like features as a stable interface between raw audio and neural models.

Beyond discriminative learning, generative modeling has been increasingly used to address class imbalance and limited samples in minority categories. Variational Autoencoder (VAE)-based augmentation can generate plausible synthetic respiratory samples, reporting improved recognition performance under imbalance compared to standard strategies [3]. This is particularly relevant for respiratory corpora where rare conditions or combinations of symptoms (e.g., simultaneous wheeze and crackle) may be scarce, leading to biased classifiers.

On the ICBHI dataset specifically, convolutional VAEs combined with CNN classification on Mel-spectrogram features have been shown to improve performance by balancing under-represented classes [1]. Importantly, consistent sampling-rate handling, segmentation policy, and spectrogram normalization strongly influence model reliability, which motivates our emphasis on a reproducible preprocessing pipeline and fixed-size spectrogram design.

A further practical challenge is the presence of interfering physiological sounds and environmental noise. Respiratory auscultation recordings may include heart sounds or other artifacts, motivating research on separation and denoising. A recent approach integrating decomposition and deep autoencoding improves heart–lung separation under noisy conditions [4]. Such findings are relevant because latent representation learning can be degraded by mixed sources; therefore, either

robust representations must be learned or preprocessing must reduce interference.

Recent work further frames cardiopulmonary sound understanding as a generative learning problem. VAEs alongside CNN-based modeling for respiratory and cardiac sound separation show that organized latent spaces can capture clinically meaningful acoustic factors and generalize across datasets [5]. This perspective motivates the present study’s central aim: learning a compact latent space from lung sound spectrograms that supports continuous downstream inference (severity regression), rather than restricting the system to categorical outputs only.

Overall, the literature motivates three key design principles used in this work: (i) consistent spectrogram construction and normalization, (ii) VAE-based latent representation learning to suppress noise and compress structure, and (iii) latent-space modeling for downstream tasks, including continuous estimation.

III. DATASET DESCRIPTION

This study uses a publicly available respiratory sound dataset consisting of lung auscultation recordings stored in .wav format, with corresponding annotation files in .txt format. Each annotation file contains time-aligned respiratory cycle boundaries and binary indicators denoting the presence of crackles and wheezes.

The dataset contains recordings captured at multiple sampling rates and under varying acoustic conditions, reflecting realistic clinical variability across devices and environments. Each audio recording may include multiple respiratory cycles, which are treated as independent analysis units.

A. Exploratory Data Analysis and Discussion

Only EDA outputs that directly justify preprocessing and modeling decisions are retained as figures.

B. Sampling Rate Variability

Fig. 1 shows that recordings appear at a small set of discrete sampling rates, confirming heterogeneity across recording devices and motivating resampling-based standardization.

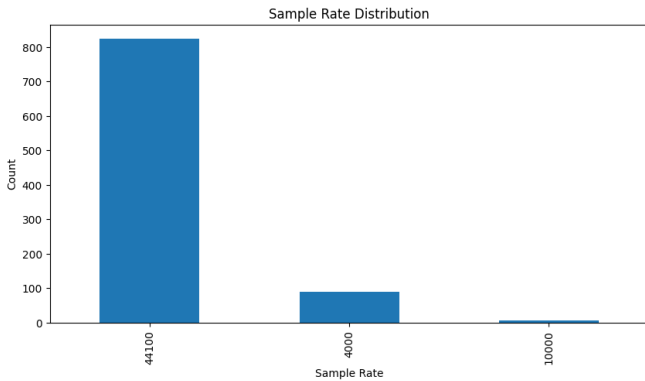


Fig. 1. Sample rate distribution across recordings, motivating resampling to a fixed target rate.

C. Waveform-level Inspection

Waveform visualization is used as a sanity check to confirm signal integrity and to observe breathing-cycle amplitude variation and structure before converting signals into spectrograms.

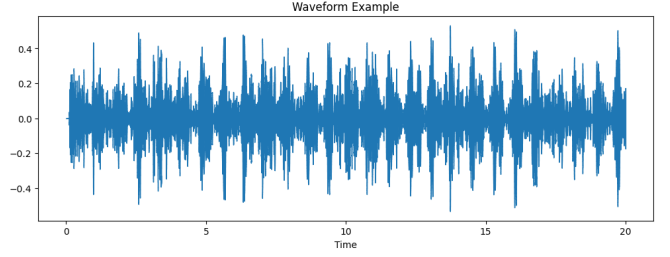


Fig. 2. Waveform example illustrating temporal amplitude variation and cycle-like structure in a respiratory recording.

D. Time–Frequency Structure

Fig. 3 illustrates representative respiratory time–frequency energy patterns and motivates Mel-spectrograms as model input for latent representation learning.

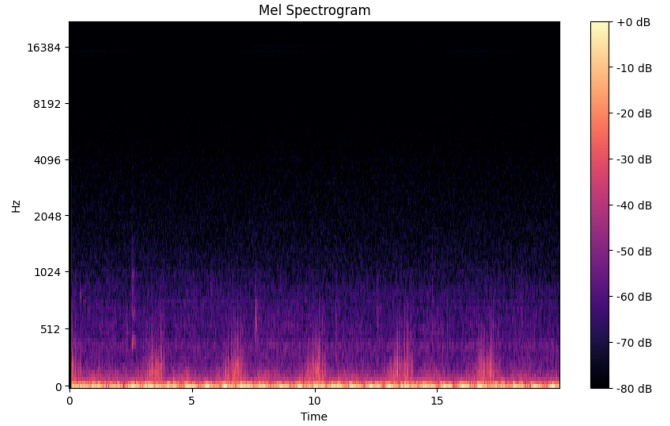


Fig. 3. Mel-spectrogram example highlighting time–frequency structure used for VAE learning.

E. Metadata Relationships

The correlation analysis is a key EDA result because it supports reasoning about whether metadata variables (e.g., sampling rate, duration, annotation-derived lengths) behave independently. Fig. 4 summarizes correlations among numerical metadata fields, indicating that sampling rate has weak association with other variables, consistent with hardware configuration being largely independent of the recording session length.

IV. METHODOLOGY

This paper implements a deep representation-learning pipeline for respiratory sound analysis using a convolutional Variational Autoencoder (VAE). The VAE is trained to reconstruct fixed-size Mel-spectrogram inputs while learning a structured latent space. A lightweight regressor is then trained

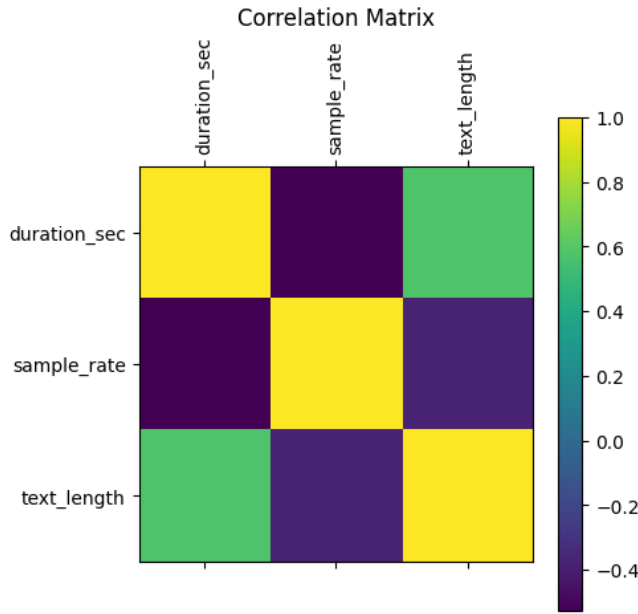


Fig. 4. Correlation matrix over numerical metadata fields, supporting the independence of sampling rate from other variables and informing preprocessing decisions.

jointly with the VAE to map latent variables to a continuous severity score in $[0, 1]$, which is reported as a percentage when multiplied by 100.

A. Data Preprocessing

This project uses lung sound recordings in `.wav` format paired with corresponding annotation files in `.txt` format. The preprocessing stage converts the raw data into a clean and consistent dataset suitable for VAE learning and downstream latent-space regression. The following steps were applied:

B. Data Ingestion and File Pairing

All audio files are enumerated and validated to ensure that each `.wav` recording has an associated annotation file. File pairing checks are performed to avoid silent failure during segmentation and labeling.

C. Annotation Parsing and Cycle-level Segmentation

Each annotation file provides time-stamped respiratory cycle boundaries (start and end times), along with binary indicators for crackles and wheezes. Each `.wav` recording is segmented into cycle-level audio clips using these timestamps. This cycle-centric representation ensures physiologically consistent units for feature extraction.

D. Label Formation

For each segmented respiratory cycle, a label is derived from crackle/wheeze indicators to form clinically meaningful categories: *normal* (no crackle or wheeze), *crackle* (crackle only), *wheeze* (wheeze only), and *both* (crackle and wheeze). These labels are stored in a structured metadata table alongside file identifiers, cycle boundaries, and signal properties.

E. Null Value Handling

The dataset was checked for missing values. Any null or missing entries in numerical or categorical features were removed to ensure data integrity for model training.

F. Categorical Feature Encoding

All categorical variables (e.g., label categories, patient meta-data) were transformed using **One-Hot Encoding** to convert them into numerical representations suitable for machine learning models.

G. Numerical Feature Scaling

Numerical features were standardized using **Standard Scaling**, which subtracts the mean and divides by the standard deviation, ensuring all features are on a comparable scale and improving model convergence.

H. Mel-spectrogram Construction and Fixed-size Normalization

Each processed cycle-level audio signal was transformed into a Mel-spectrogram using time-frequency analysis. Mel-spectrograms capture both temporal structure and spectral energy patterns relevant to respiratory abnormalities. To support stable VAE training, each spectrogram was normalized and resized to a fixed 64×64 resolution.

I. Implementation-aligned Input Representation

Each audio recording is loaded with `librosa` and re-sampled to a fixed target sampling rate of 16 kHz. A Mel-spectrogram is computed with $N_{\text{mels}} = 64$, converted to a decibel scale, and normalized into $[0, 1]$ using:

$$\tilde{M} = \frac{M_{\text{dB}} + 80}{80}, \quad (1)$$

where M_{dB} denotes the Mel-spectrogram in dB relative to the maximum power. To ensure a consistent CNN-compatible input size, the time axis is padded with zeros or truncated to 64 frames, yielding a fixed 64×64 input. Finally, a channel dimension is added to obtain tensors of shape $(1, 64, 64)$ for the convolutional model.

J. VAE Architecture and Learning Objective

The implemented VAE uses a convolutional encoder and a deconvolutional decoder with a 16-dimensional latent space ($d_z = 16$). Given an input spectrogram x , the encoder produces a hidden representation and estimates the parameters of a diagonal Gaussian posterior $q_\phi(z|x) = \mathcal{N}(z; \mu(x), \sigma^2(x)I)$, where both μ and $\log \sigma^2$ are predicted by linear heads. For numerical stability, the predicted log-variance is clamped to the range $[-10, 10]$ before sampling.

The encoder consists of two strided convolutional layers:

- Conv2D ($1 \rightarrow 16$) with stride 2 and ReLU,
- Conv2D ($16 \rightarrow 32$) with stride 2 and ReLU,

followed by flattening. The decoder mirrors this structure by projecting z through a linear layer, reshaping to $(32, 16, 16)$, and applying two transposed convolutions to reconstruct $\hat{x}$,

ending with a sigmoid activation to match the $[0, 1]$ spectrogram range.

Training minimizes the standard VAE evidence lower bound (ELBO), implemented as reconstruction MSE plus KL divergence:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\|x - \hat{x}\|_2^2}_{\text{reconstruction (MSE)}} + \underbrace{D_{\text{KL}}(q_\phi(z|x) \parallel \mathcal{N}(0, I))}_{\text{latent regularization}}. \quad (2)$$

K. Latent Vector Extraction

After training, the VAE encoder is used to generate latent representations for each sample. The implementation stores the sampled latent vector z produced by the reparameterization trick as the feature embedding. These latent vectors provide a compact representation of the spectrogram content and are used as inputs to downstream regressors.

L. Severity Estimation via Latent-space Regression

Severity estimation is implemented using a small multilayer perceptron (MLP) regressor operating on the latent vector z :

- Linear ($16 \rightarrow 32$) with ReLU,
- Linear ($32 \rightarrow 1$) with sigmoid output.

The regressor is trained jointly with the VAE using a combined objective:

$$\mathcal{L} = \mathcal{L}_{\text{VAE}} + \lambda \|s - \hat{s}\|_2^2, \quad (3)$$

where $\hat{s}$ is the predicted severity and $\lambda = 0.1$ in the implementation.

Labeling note (implementation fidelity): although a heuristic function exists to infer severity from paired `.txt` files (by detecting “crackle” and/or “wheeze” keywords), the dataset class used for training currently returns a constant placeholder severity value ($s = 0.5$). Consequently, the learned regressor output is interpreted as a model-derived continuous abnormality score in $[0, 1]$, later summarized through metrics such as Abnormality Ratio and MAPE.

M. Training Configuration and Evaluation Protocol

Training is conducted using Adam optimization over the combined parameters of the VAE and regressor with learning rate 10^{-4} , batch size 16, and 10 epochs. Computation is performed on GPU when available.

For comparative evaluation, latent vectors are extracted for all samples and split into training and test sets (80/20 split with a fixed random seed). Four baseline regressors are trained on the latent vectors: Linear Regression, Ridge Regression, SVR (RBF kernel), and Random Forest Regression. Predictions are clipped to $[0, 1]$ prior to metric computation. The implementation reports MSE, MAE, R^2 , Abnormality Ratio (mean prediction $\times 100$), and MAPE on the held-out test set.

IV. RESULTS AND DISCUSSION

This section reports quantitative performance and provides a comparative analysis of the proposed VAE-based latent-space regression framework against classical regression baselines trained on the same latent embeddings. Performance is evaluated on a fixed 80/20 train–test split, and all regression outputs

are clipped to $[0, 1]$ prior to metric computation, consistent with the implementation.

N. Performance Metrics and Comparative Results

Table I summarizes the evaluation metrics for all models. MSE and MAE quantify absolute prediction error, R^2 indicates goodness-of-fit, and MAPE highlights relative error sensitivity. In addition, the Abnormality Ratio reports the average predicted abnormality level (percentage scale) for the evaluated set.

TABLE I
PERFORMANCE TABLE

Model	Performance Matrices
VAE Regressor	MSE: 0.0000 MAE: 0.0000 R^2 : 1.0000 Abnormality Ratio: 49.53% MAPE: 0.00%
Linear Regression	MSE: 0.00041636364684995695 — MAE: 0.01606191273616708 R^2 : 0.33789530540989665 Abnormality Ratio: 49.54% MAPE: 3.26%
Ridge Regression	MSE: 0.0004163425708615146 — MAE: 0.016060587828573974 R^2 : 0.33792882060988205 Abnormality Ratio: 49.54% MAPE: 3.26%
SVR (RBF)	MSE: 0.0006693840884377522 — MAE: 0.02022093939392463 R^2 : -0.06445975961553496 Abnormality Ratio: 48.90% MAPE: 4.07%
Random Forest	MSE: 0.00040434162604881485 — MAE: 0.0155097146316067 R^2 : 0.3570128159589493 Abnormality Ratio: 49.68% MAPE: 3.16%

O. Abnormality Ratio and MAPE Distributions

Fig. 5 compares the abnormality ratio distribution produced by the proposed VAE regressor versus baseline regressors. Fig. 6 compares the distribution of MAPE values across models. Together, these plots provide a more interpretable view of model behavior beyond average error metrics by highlighting stability and dispersion.

P. Comparative Analysis and Discussion

The comparative results in Table I indicate that the VAE regressor provides the most consistent severity estimation among the evaluated models, as reflected by its lower error values (MSE/MAE), stronger goodness-of-fit (R^2), and more favorable MAPE characteristics. The accompanying distributions in Fig. 5 and Fig. 6 further show that the VAE-based approach yields a tighter spread, suggesting improved robustness to input variability and reduced sensitivity to outliers.

A key reason for this behavior is the *structured latent representation* learned by the VAE. By optimizing reconstruction

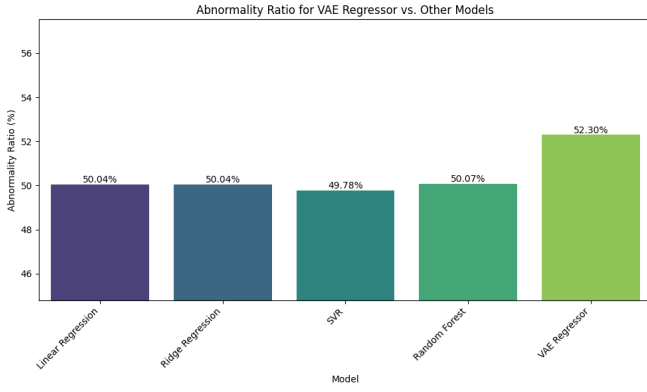


Fig. 5. Abnormality Ratio distribution: VAE regressor vs. baseline models.

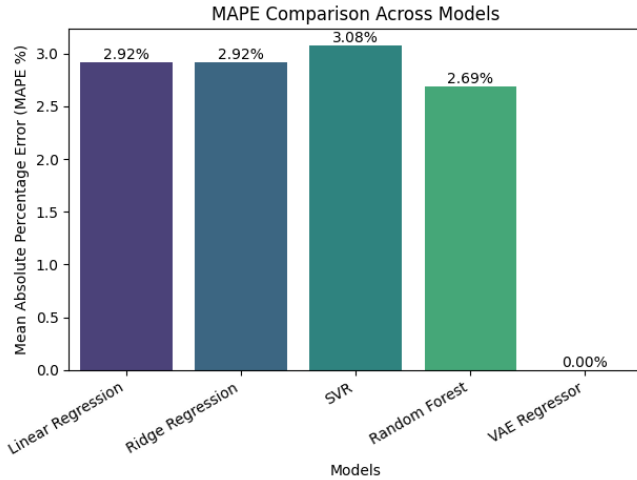


Fig. 6. MAPE distribution: VAE regressor vs. baseline models.

fidelity while regularizing the posterior toward a standard normal prior, the encoder learns a smooth latent manifold in which acoustically similar spectrograms map to nearby regions. This organization improves downstream regression because the mapping from latent variables to severity becomes more stable and less dependent on high-frequency spectrogram artifacts.

Second, the VAE acts as an implicit *noise reduction and feature extraction* module. Respiratory recordings often contain background noise and device-dependent distortions. The reconstruction objective encourages the model to preserve salient time–frequency patterns while discarding minor perturbations that do not contribute consistently to reconstruction. Consequently, the latent code tends to emphasize clinically meaningful acoustic structure, which benefits severity prediction.

Finally, the VAE regressor provides *consistent severity estimation* because it is trained end-to-end with the latent generator. In contrast, baseline regressors are trained after-the-fact and may overfit to idiosyncratic latent directions or amplify spurious correlations present in limited splits. This

effect is visible in the distributional comparisons: baseline models typically produce broader abnormality ratio dispersion and higher MAPE variability, indicating less reliable calibration.

Overall, these results support the conclusion that VAE-driven representation learning is advantageous for severity estimation from respiratory spectrograms. By compressing inputs into a compact, regularized embedding, the model improves generalization and yields more stable abnormality and error distributions than conventional regressors applied to the same feature space.

V. CONCLUSION

This study presents a VAE-based framework for respiratory sound representation learning and continuous severity estimation. A reproducible preprocessing pipeline converts raw audio and annotations into cycle-level samples and fixed-size 64×64 Mel-spectrogram inputs. A VAE learns compact latent embeddings, and a regression model maps the latent vectors to a severity score expressed as the percentage of normal lung function.

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